

ASMI JAIN

Software Engineer | Backend Engineer | Full-Stack Developer | Immediate Joiner

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PROFESSIONAL SUMMARY

Software Engineer with ~2 years of experience developing and maintaining enterprise-grade desktop applications using **C#, C++, .NET Framework, WPF, MVVM, and MFC**. Improved document-processing performance by ~85% through **multithreading (TPL)** and **asynchronous programming** in a production environment. Experienced in production support, performance optimization, concurrency, debugging, and enterprise application maintenance. Hands-on experience in **backend** development with **FastAPI, JWT Authentication, SQLAlchemy, PostgreSQL**, and **React**-based frontend development. **SIH 2022 National Winner** and strong foundations in Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, and System Design. **Immediate Joiner**.

TECHNICAL SKILLS

Languages: C#, C++, Java, Python, JavaScript, SQL

Backend: REST APIs, FastAPI, JWT Authentication, SQLAlchemy

Frontend: React.js, HTML5, CSS3, JavaScript

Frameworks: WPF, MVVM, .NET Framework, Visual Studio

Databases: PostgreSQL, Firebase Realtime Database

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, System Design

AI / ML: LangChain, Groq API, TensorFlow, OpenCV, YOLOv3

Tools & Cloud: Git, GitHub, Perforce, Azure

WORK EXPERIENCE

Software Engineer — Celstream Technologies | Aug 2024 – May 2026 | Bengaluru

- Performance Optimization:** Improved enterprise document-processing performance by ~85% through redesigning **asynchronous** data binding and implementing **multithreading** using **C# Task Parallel Library (TPL)**, significantly reducing processing latency and improving application responsiveness.
- Scalable UI:** Developed desktop application features using **C#, .NET Framework, WPF**, and **MVVM** by building reusable UI components, dynamic DataGrid loading, eliminating UI freezes for **datasets** exceeding **1,000+** rows in production enterprise deployments, responsive layouts, and workflow enhancements.
- Maintained and enhanced **legacy C++/MFC** modules by implementing Page Setup workflows, toolbar enhancements, UI state synchronization, access-control logic, and production bug fixes using Xtreme Toolkit Pro (XTP).
- Concurrency & Reliability:** Improved application reliability by implementing **file-locking**, **retry mechanisms**, and **state-management** logic, preventing concurrent access issues and reducing the risk of document corruption.
- Resolved production issues by **performing root cause analysis (RCA)**, debugging complex C++ and C# application workflows, analysing logs, and delivering reliable fixes for enterprise customers as part of L3 production support.
- Collaborated with senior engineers on feature development, code reviews, production support, and defect resolution within a large enterprise C++/.NET codebase using **Agile** development practices.

Project Trainee (Internship) — Celstream Technologies | Feb 2024 – Aug 2024 | Bengaluru

- Onboarded onto a 100K+ line enterprise **C++/C#/WPF codebase** and implementing production features, debugging application issues, and enhancing existing functionality under senior engineer mentorship.
- Acquired hands-on experience with **WPF, MVVM, MFC, Perforce, Visual Studio**, debugging large codebases, and **Agile** software development practices.

PROJECT

Make Your Own Adventure Story AI | [live-link](#) | [GitHub Code link](#)

React • FastAPI • JWT • LangChain • PostgreSQL • SQLAlchemy • Groq • Vercel • Render

- AI Story Generation:** Architected an AI-powered interactive storytelling platform using FastAPI, React, PostgreSQL, LangChain, and Groq LLM to generate dynamic, user-driven narratives.
- Authentication:** Implemented secure **JWT-based authentication** using **FastAPI, SQLAlchemy**, password hashing, and role-based authorization to protect **RESTful APIs**.
- Backend:** Developed scalable backend services using **FastAPI, SQLAlchemy**, and **PostgreSQL** for user management, AI story persistence, and **API** communication.
- AI Integration:** Integrated **LangChain** with **Groq LLM** to generate dynamic, choice-driven interactive stories using recursive prompt engineering and structured workflows.
- Architecture:** Improved **API** reliability by implementing **Pydantic** validation, structured request models, and centralized exception handling.
- Frontend:** Developed a responsive **React** frontend with **authenticated** routing, **API** integration, and dynamic rendering of **AI-generated** interactive stories.

- Moving Vehicle Registration Plate Detection** | *Python • YOLOv3 • TensorFlow • OpenCV • Tesseract OCR • Firebase Realtime Database*
Smart India Hackathon 2022 — National Winner, Ministry of Coal, Govt. of India
- Engineered a computer vision pipeline using **YOLOv3, TensorFlow, OpenCV, and OCR** to automate vehicle number plate recognition under varying environmental conditions.
 - Built **Firebase real-time Database** tracking and analytics for monitoring vehicle movement and operational efficiency.
 - Optimized object detection performance through dataset preprocessing and custom **YOLOv3** model fine-tuning for challenging real-world scenarios.

EDUCATION

B.E. Computer Science & Engineering — BNM Institute of Technology, Bangalore | *2021 – 2024 · CGPA 8.5/10*
Std. XII (Science) — Sophia Girls Secondary School, Bhilwara | *2018 – 2020 · 90%*

ACHIEVEMENTS

- Hackathon 5.0 Runner-Up — IEEE-BNMIT Student Branch (2021); built a Mental Health App.
- Secretary, Student Council (12th Std.) — led a 12-member team across student governance and events.